

Motion Analysis Mid-Term Assignment

Objective:

Track object movement over time using motion
analysis on video input

original vs result

- Frame in the original video
- Frame in the result



Avg background

Estimate the background by averaging every 10 frames



change

- We compare each frame of the video with the averaged background in grayscale, and apply a pixel-wise threshold: differences above the threshold are marked as white (foreground), otherwise black (background).
- The idea is that pixels that remain similar to the averaged background are considered static and part of the background, while significant pixel changes indicate motion and are classified as foreground.

change



Threshold

- We select a threshold that controls sensitivity: a low threshold detects small movements but increases noise, while a high threshold reduces noise but may miss subtle motion, so a balance is required.

— t-5



t-50



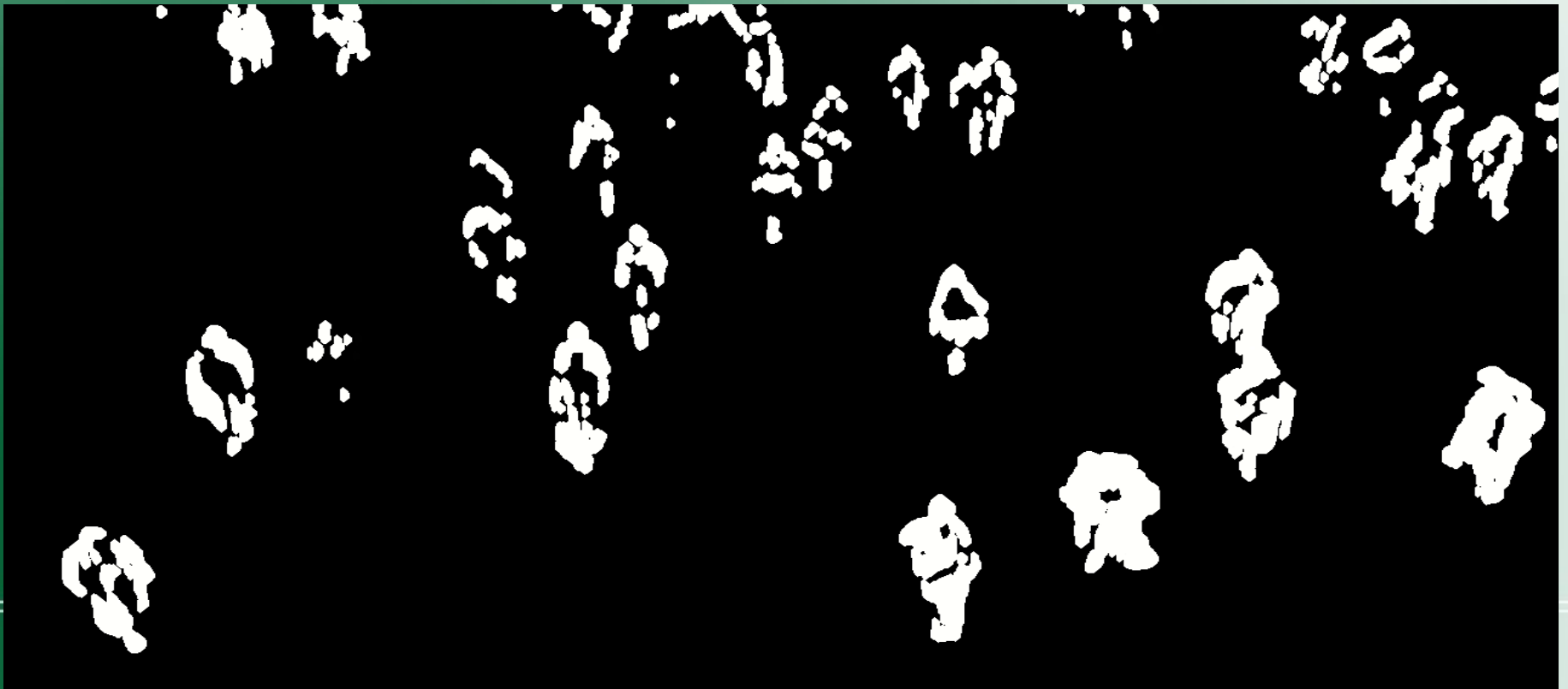
OTSU Threshold

Using Otsu's Method, the best threshold is selected automatically, avoiding the need to manually test different threshold values.



Morphological Operations

- We apply morphological operations: erosion to reduce noise in the binary mask, followed by dilation to reconnect nearby regions .



Tracking

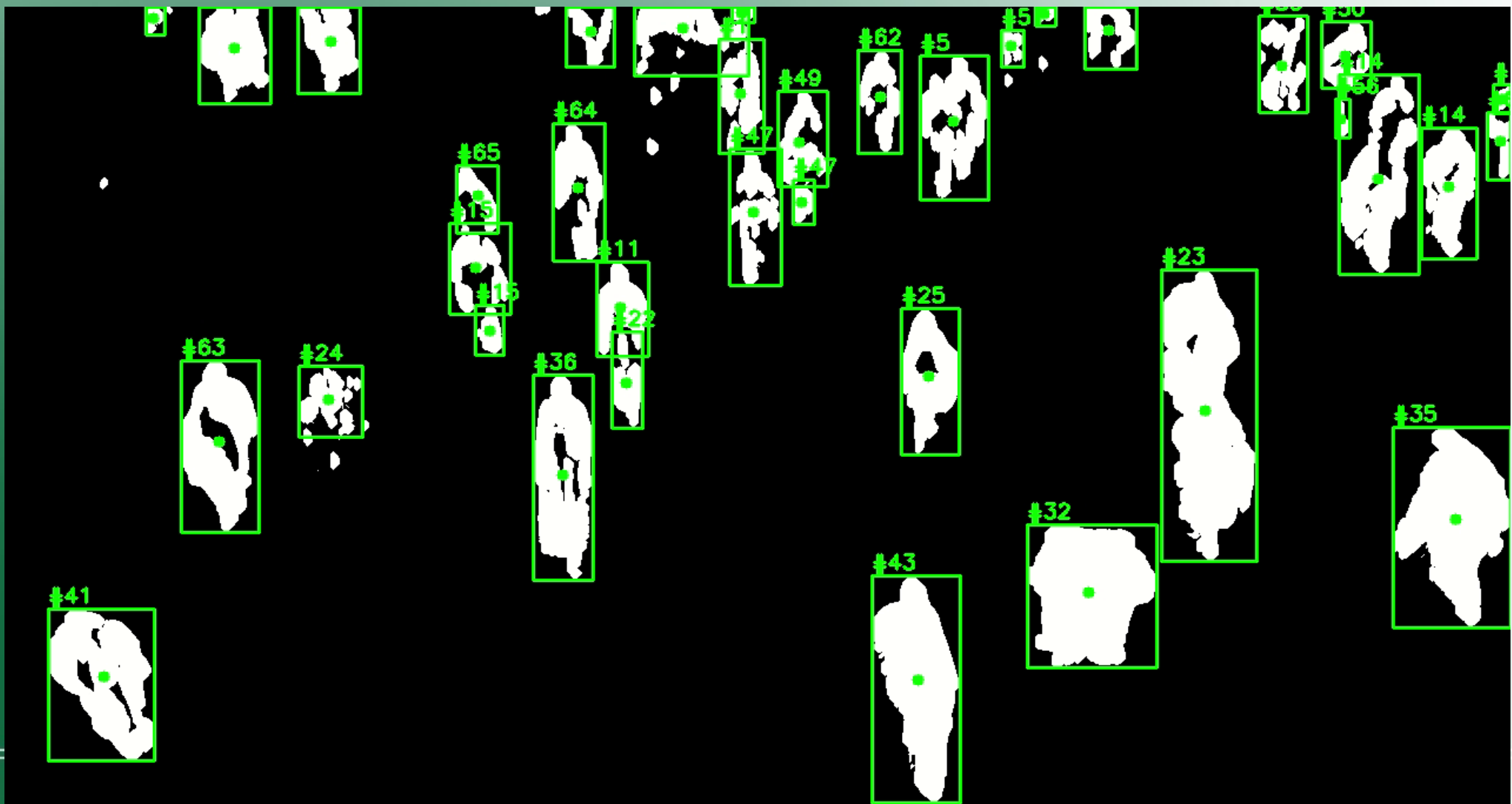
We will perform object tracking. First, we average changes over time in a decay-like manner, so regions with consistent motion become more prominent, while random noise gradually fades out.

- Then we connect the remaining white pixels to form coherent motion regions (blobs), grouping nearby detections into meaningful objects.
- last, each detected blob is enclosed within a bounding box, which represents the final tracked object in the scene.

Tracking over time

- Finally, we perform temporal object tracking. For each blob, we compute its centroid and assign an ID. In each new frame, we compare the current centroids with the previous ones: if a nearby match is found, the same ID is propagated to maintain object continuity; otherwise, a new ID is created for a newly detected object. Each tracked object is then represented by a bounding box, annotated with its corresponding centroid ID.

Tracking



final

- Last, we draw the bounding boxes over the original video to visualize the tracked objects.

